

GraphCL-DTA: a graph contrastive learning with molecular semantics for drug-target binding affinity prediction

Xinxing Yang, Genke Yang, and Jian Chu

Abstract—Drug-target binding affinity prediction plays an important role in the early stages of drug discovery, which can infer the strength of interactions between new drugs and new targets. However, the performance of previous computational models is limited by the following drawbacks. The learning of drug representation relies only on supervised data without considering the information in the molecular graph itself. Moreover, most previous studies tended to design complicated representation learning modules, while uniformity used to measure representation quality is ignored. In this study, we propose GraphCL-DTA, a graph contrastive learning with molecular semantics for drug-target binding affinity prediction. This graph contrastive learning framework replaces the dropout-based data augmentation strategy by performing data augmentation in the embedding space, thereby better preserving the semantic information of the molecular graph. A more essential and effective drug representation can be learned through this graph contrastive framework without additional supervised data. Next, we design a new loss function that can be directly used to adjust the uniformity of drug and target representations. By directly optimizing the uniformity of representations, the representation quality of drugs and targets can be improved. The effectiveness of the above innovative elements is verified on two real datasets, KIBA and Davis. Compared with the GraphDTA model, the relative improvement of the GraphCL-DTA model on the two datasets is 2.7% and 4.5%. The graph contrastive learning framework and uniformity function in the GraphCL-DTA model can be embedded into other computational models as independent modules to improve their generalization capability.

Index Terms—Drug-target binding affinity prediction, Graph contrastive learning, Molecular graph, Uniformity of representations, Drug discovery

I. INTRODUCTION

DEVELOPING a new drug for a specific disease requires an estimated cost of 260 million US dollars and about 17 years to pass the review of the Food and Drug Administration (FDA) [1], [2]. Small molecule drugs, as compounds, can cure related diseases by binding to specific targets. Therefore, predicting binding affinity between drugs and targets is a crucial task that can accelerate drug discovery [3].

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Previous studies used high-throughput screening techniques to study the binding affinity between drugs and targets [4]. This exhaustive method is not only expensive but also time-consuming. Moreover, this method lacks theoretical feasibility as millions of compounds currently known to humans and hundreds of targets have therapeutic effects on diseases, the permutation and combination of which are excessive. Therefore, the development of high-precision computational models [5] for predicting the binding affinity between drugs and targets can reduce the cost of identifying interactions between drugs and targets and increase the speed, which is of great significance for promoting drug discovery.

The computational models [6]–[10] use the information of the drug and the target as the input, and the output is the half maximal inhibitory concentration (IC₅₀) after the drug binds to the target. The binding affinity between the drug and the target reflects their interaction strength, which can provide more information on the relationship between the drug and the target. Computational models generally consist of three modules: drug representation learning module, target representation learning module, and prediction module. For example, Ozturk et al. [11] proposed the DeepDTA model, using the sequence information of drugs and proteins to predict their binding affinity. DeepDTA uses the convolutional neural network (CNN) as a representation learning module for drugs and targets. They used simplified molecular input line entry system (SMILES) strings and protein sequences as raw input. Then, the drug and target representations are concatenated into the fully connected neural network to obtain the predicted binding affinity. Based on DeepDTA, Abbasi et al. [12] proposed the DeepCDA model. The representation learning module of this model combines CNN and a long short-term memory network (LSTM). The workflow first inputs the drug and target sequence information into CNN and LSTM sequentially to capture the global information among the sequence information. Then, an attention mechanism was used in their prediction module to infer the importance of weights in sequences.

Similarly, Zeng et al. [13] proposed the MATT-DTI model, using the SMILES strings as the input of the drug representation learning module and the protein sequence as the input of the target representation learning module. The prediction module adopts a multi-head attention mechanism and multi-layer neural network. In the drug representation learning module, they proposed a relation-aware self-attention model for learning

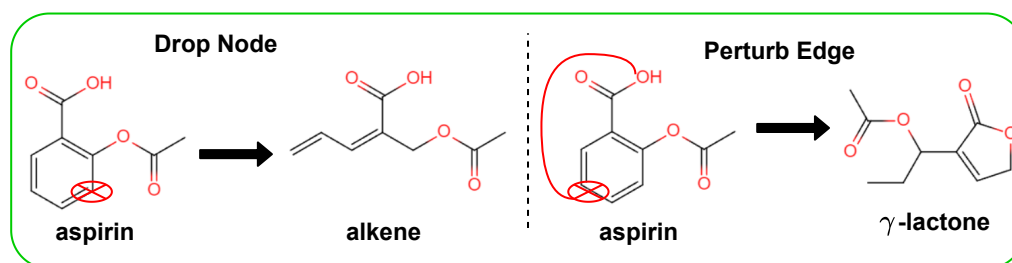


Fig. 1. Dropout-based augmentation strategies destroy molecular semantics.

drug representations, which could capture the correlation of atoms in drugs. In the target representation learning module, they employed a one-dimensional convolutional neural network (1D CNN) to learn the representation of the target. Then, the drug and target representations were concatenated and input into the prediction module to obtain the value of the predicted binding affinity.

Representing drugs with SMILES strings is not a natural way. Therefore, Nguyen et al. [14] proposed the GraphDTA model and designed a drug representation learning module based on molecular graphs. The drug representation learning module uses the graph convolutional network (GCN) to learn the structural information from the molecular graph. Then, it obtains the drug representation through the global max pooling operation. Likewise, this model learns target representations using protein sequences and 1D CNN. In the prediction module, they concatenated the drug and target representations and input them into the fully connected neural network to obtain the predicted binding affinity. Validated on multiple datasets, drug representations learned using molecular graphs have better generalization ability than SMILES strings. Bidirectional LSTM has demonstrated its superiority over 1D CNN in capturing the characteristics of temporal signals in many fields. Therefore, based on GraphDTA, Mukherjee et al. [15] proposed a new drug-target binding affinity prediction model. The difference is that this model uses a bidirectional LSTM model as the target representation learning module. The experimental results showed that the performance of the bidirectional LSTM model is better than that of 1D CNN. Li et al. [16] improved the prediction module and proposed a two-way attention neural network for the fusion of drug and target representations. This approach enables the model to focus on the effective sites of nuclear amino acids and makes itself interpretable.

However, the previous work has the following two limitations. Firstly, the above methods input the molecular graph into the GCN [17]–[19], and use the supervised data of known drug-target binding affinity to inversely optimize the parameters in the GCN to learn the drug representation. The disadvantage is that it relies on a large amount of supervised data to learn the appropriate drug representation, which is often time-consuming and expensive. Moreover, such drug representation is dependent on the supervised data. The drug's characteristics cannot be described in essence without the molecular graph data.

Graph contrastive learning (GCL) [20], as a self-supervised framework, can learn high-level intrinsic features from the drug molecular graph to distinguish drugs without supervised

data. However, previous data augmentation strategies mainly use random deletion of edges or nodes to generate different views, which changes the semantics of molecular graphs and cannot benefit from the relational inductive bias of molecular graphs. As shown in Figure 1, the deletion of nodes and edges on the molecular graph of aspirin produces molecules with other properties [21].

Secondly, the above models use Mean Square Error (MSE) as the loss function for parameter optimization. This loss function does not consider the uniformity of drug and target representation. At the same time, in the field of recommender systems, studies have shown that the uniformity of representation can improve the model's performance [22], [23]. Previous computational models focused on designing effective representation learning and prediction modules. However, few studies have been conducted to improve the loss function of drug-target binding affinity.

Therefore, in response to the above limitations, inspired by graph contrastive learning [24] and representation uniformity [25], we propose GraphCL-DTA, a graph contrastive learning with molecular semantics for drug-target binding affinity prediction. The GraphCL-DTA model consists of a drug representation learning module, a target representation learning module, and a prediction module. In the drug representation learning module, we propose a graph contrastive learning framework that can better preserves molecular semantics. The framework flow is as follows: 1. Input the molecular graph of the drug into the GCN to obtain the drug representation. 2. Add different random noises to the drug representation to generate two contrastive views. 3. Optimize the contrastive views so that the views of the same drug become closer in the representation space and views of different drugs become farther apart. The advantage of this framework is that complex data augmentation strategies are no longer needed to generate contrastive views. Contrastive views can be created by adding controllable random noise directly in the representation space of drugs, resulting in variance across different contrasted views. By controlling the magnitude of random noises, the semantics of molecular graphs are better preserved. A more essential and effective drug representation can be learned through this graph contrastive framework without additional supervised data.

In the target representation, we use the same operation as GraphDTA: input the protein sequence into the 1D CNN to obtain the target representation. In the prediction module, we concatenate the drug and target representations to form a one-dimensional vector, which is then input into a multi-layer

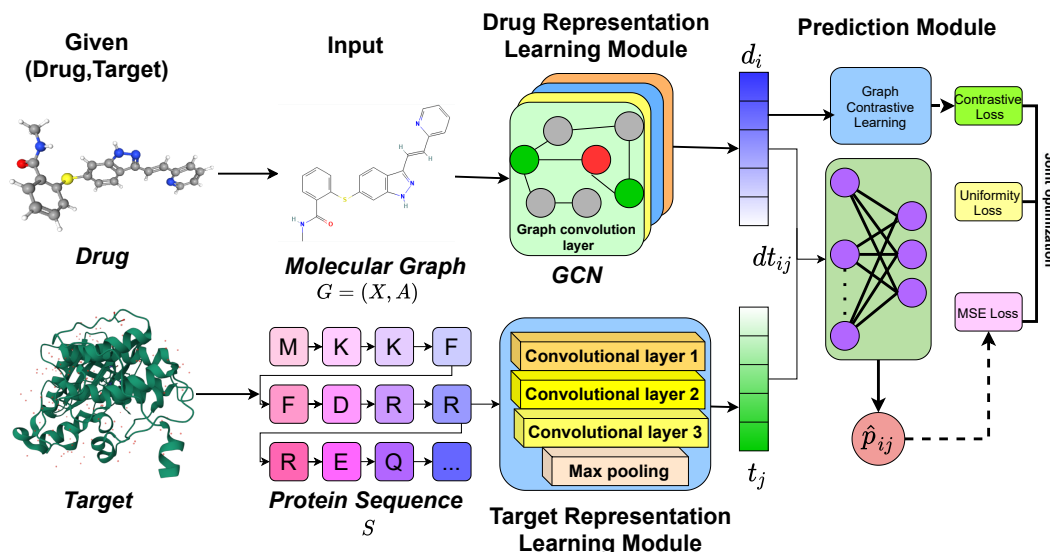


Fig. 2. The architecture of our proposed GraphCL-DTA model.

fully connected neural network to obtain the predicted binding affinity.

In addition, we design a new loss function that can be directly used to adjust the uniformity of drug representation and target representation smoothly. By minimizing this loss function, the uniformity and quality of drug representation and target representation are enhanced, thereby improving the predictive ability of the GraphDTA model.

In general, the main contributions of this work are as follows:

- 1) We design a graph contrastive learning framework for learning drug representations. Semantic information of molecular graphs can be preserved without data augmentation strategies. A more essential and effective drug representation can be learned through this graph contrastive framework without additional supervised data.
- 2) We design a loss function optimizing the uniformity of drug and target representations to improve the representation quality.
- 3) Extensive experiments have been conducted on two publicly available datasets. The results demonstrate the effectiveness of the graph contrastive learning framework and the loss function optimized for uniformity.

The rest of the paper is organized as follows. Section 2 introduces our proposed GraphCL-DTA model. Section 3 introduces the relevant experimental results. Section 4 makes the conclusion.

II. METHOD

Figure 2 shows the architecture of our proposed GraphCL-DTA model. The GraphCL-DTA model consists of three modules: the drug representation learning module, the target representation learning module, and the prediction module. The workflow of the GraphCL-DTA model, simply put, is that the molecular graph of the drug is input into the drug representation learning module to obtain the representation of the drug, and the protein sequence of the target is input into the target representation learning module to obtain the

representation of the target. The drug and target representation are concatenated and then input into the prediction module to obtain the predicted binding affinity.

The drug representation learning module is the focus of this work, in which we design a graph contrastive learning framework for molecular graphs, which can learn better drug representation without complicated manual augmentation strategies and additional supervised data. The target representation learning module is similar to GraphDTA, which uses a 1D CNN. The prediction module uses a fully connected neural network. Another focus of this work is a new loss function for directly optimizing the uniformity of the representation, thereby improving the quality of the drugs and targets representation. Firstly, we illustrate how to use GCN to learn drug representations. Then, we sequentially introduce the target representation learning module, prediction module, and optimization function. Next, we explain how to use the graph contrastive framework to improve the representation quality of drugs. The final part is the loss function optimized for uniformity.

A. Drug representation using GCN

Given a drug i , its molecular graph can be defined as $G = (X, A)$. $X \in \mathbb{R}^{n \times k}$ represents the features matrix of atoms. n represents the number of atoms. k represents the feature number of each atom. $A \in \mathbb{R}^{n \times n}$ represents the adjacency matrix. If $A[i][j] = 1$, a bond is connected between atom i and atom j . Otherwise, there is no bond connection. The GCN takes $G = (X, A)$ as input and outputs vector d_i , which is the representation of drug i . The propagation rules of each layer in GCN are shown in equation (2).

$$Z^{l+1} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} Z^l W^l) \quad (1)$$

$\tilde{A} = A + I$, I represents the identity matrix. This operation aims to add a self-connection to the adjacency matrix. $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Z^l represents the input of the l th layer. Note that $Z^0 = X$. W^l represents the trainable weight matrix of the l

layer. We define the output of the last layer of the GCN as Z^F , which is the representation of the node level. We need the representation of the graph level, so we use the equation (2) for Z^F to do a global max pooling operation.

$$d_i = GMP(Z^F) \quad (2)$$

Among them, GMP represents the global max pooling layer, and d_i represents the drug i . The drug representation learning module in this section is similar to the corresponding module in the GraphDTA model. In the actual code, the vector d_i can subsequently be connected to multiple fully connected layers to further improve the representation quality.

B. Target representation using 1D CNN

We adopt the same approach as GraphDTA to learn the representation of targets. The protein sequence of the target is input into three 1D CNN layers and a max pooling layer to obtain the representation vector of the target. The protein sequence of the target uses a string of ASCII characters representing amino acids. We use integer encoding to convert it into an integer sequence vector (Alanine (A) is 1, Cystine (C) is 3, and so on). Given a target j , its protein sequence is S . We input it into the target representation learning module to get its representation t_j .

C. Prediction module and optimization function

In the prediction module, we concatenate d_i and t_j into the vector dt_{ij} , and then input dt_{ij} into a fully connected neural network to obtain the predicted drug-target binding affinity $\hat{p}_{ij}$. The propagation of the fully connected function is shown in the equation (3).

$$\hat{p}_{ij} = V_3^T \sigma(V_2^T \sigma(V_1^T dt_{ij} + b_1) + b_2) + b_3 \quad (3)$$

V_1^T , V_2^T , and V_3^T are the weight parameters of each fully connected neural network. b_1 , b_2 and b_3 are bias parameters; $\hat{p}_{ij}$ represents the binding affinity between drug i and target j predicted by the GraphCL-DTA model.

$$\mathcal{L}_{mse} = \frac{1}{c} \sum_1^c (p_{ij} - \hat{p}_{ij})^2 \quad (4)$$

As shown in equation (4), we use MSE as the loss function to optimize the parameters in the model. By minimizing the $\mathcal{L}_{mse}$, the predicted value $\hat{p}_{ij}$ can continuously approach the real value p_{ij} , thereby generating a gradient, which is backpropagated to the GraphCL-DTA model for learning appropriate parameters.

D. Graph contrastive learning for drug representations

Previous dropout-based augmentation strategies change molecular graphs' semantics, which is unsuitable for drug scenarios. Therefore, we directly introduce the data augmentation strategy into the drug embedding space. This is inspired by adversarial training in the image domain and simple graph contrastive learning [24], [26], where adding controllable noise to the embedding space of an image does not change the semantic information. We add random noise to the drug

embedding space as a simple and effective augmentation strategy. The data augmentation strategy is shown in equation (5).

$$e_i^1 = d_i + \Delta_i', \quad e_i^2 = d_i + \Delta_i'' \quad (5)$$

Random noise Δ_i' and Δ_i'' are subject to two constraints. The first one is that $\|\Delta_i'\|_2 = \|\Delta_i''\|_2 = \epsilon$, and the second is that the $\Delta_i', \Delta_i'' = \tilde{\Delta} \odot \text{sign}(d_i)$, $\tilde{\Delta} \in \mathbb{R} \sim U(0, 1)$. U represents uniform distribution. The effect of the first constraint is to control the magnitude of the noise vector, which is numerically equivalent to a point on a hypersphere with radius ϵ . The second restriction ensures that the three vectors, e_i^1 , e_i^2 , and d_i are in the same hyperoctant. In this way, the contrastive views, e_i^1 and e_i^2 , will not have large semantic deviations from the original view d_i .

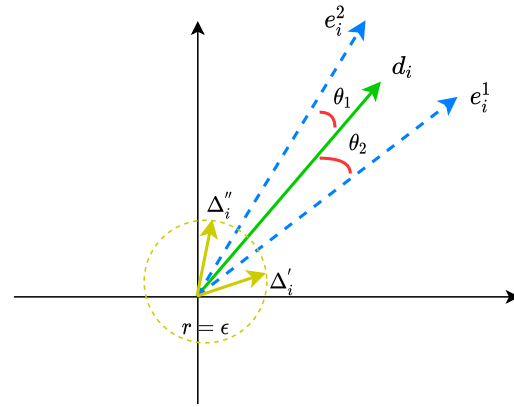


Fig. 3. An example of a data augmentation strategy in $\mathbb{R}^2$ Space.

An example of a data augmentation strategy in $\mathbb{R}^2$ Space is shown in Figure 3. The representation of the drug, d_i , is rotated by a slight angle in the embedding space under the influence of noise θ_1 and θ_2 . Each rotation angle produces an enhanced representation (e_i^1 and e_i^2) that can be directly used as a contrastive view. It is worth noting that as long as the rotation angle is small enough, the contrastive view can not only better retain the maximum effective information of the original representation, which means that the semantic information of the drug is not changed too much but also causes variance between the two representations. This makes it easier for GCL to learn information. The contrastive loss is shown below in equation (6).

$$\mathcal{L}_{gcl} = \sum_{i \in \mathcal{B}} -\log \frac{\exp(e_i^1 e_i^2 / \tau)}{\sum_{j \in \mathcal{B}} \exp(e_i^1 e_j^2 / \tau)} \quad (6)$$

Both drug i and drug j are obtained from the same batch $\mathcal{B}$. e_i^1 and e_i^2 represent two contrastive views of drug i . e_j^2 represents the contrastive view of drug j . τ represents the temperature coefficient. The effect of the contrastive loss is to strengthen the consistency of the two contrastive views belonging to the drug i . In contrast, the consistency of the contrastive views belonging to different drugs is weakened. To make graph contrastive learning effective, we jointly optimize

the contrastive loss $\mathcal{L}_{gcl}$ and the MSE loss $\mathcal{L}_{mse}$, as shown in equation (7) below. α is used to adjust the weight of the latter.

$$\mathcal{L}_{joint} = \mathcal{L}_{mse} + \alpha \mathcal{L}_{gcl} \quad (7)$$

There are two reasons why the above graph contrastive learning can improve drug representation. Firstly, through the contrastive loss $\mathcal{L}_{gcl}$, the structural information can be learned from the molecular graph without supervised data, making each drug representation highly discriminative. Secondly, the contrastive loss $\mathcal{L}_{gcl}$ can be used as a regular term in joint optimization loss $\mathcal{L}_{joint}$ to limit the complexity of the GraphCL-DTA model, remove redundant information in the representation, and thus extract its generalization ability in subsequent prediction tasks.

E. Loss function optimized for uniformity of representation

Unlike previous loss functions, we design a new version to directly optimize the uniformity of drug representation and target representation, which has been proven in data mining to improve the model's generalization performance [25]. Firstly, we measure the uniformity of drug and target representation using equations (8) and (9).

$$\mathcal{L}_{uniform}^d = \log \mathbb{E}_{u,v \sim D} e^{-2\|d_u - d_v\|_2} \quad (8)$$

$$\mathcal{L}_{uniform}^t = \log \mathbb{E}_{u,v \sim T} e^{-2\|t_u - t_v\|_2} \quad (9)$$

Taking the uniformity of drug representation as an example, we are actually computing the logarithm of the average pairwise Gaussian potential. Therefore, as shown in equation (10), we add uniformity to the MSE loss function, making the model optimize it directly. β is used to adjust the weight of the $\mathcal{L}_{uniform}^d$ and $\mathcal{L}_{uniform}^t$.

$$\mathcal{L}_{joint} = \mathcal{L}_{mse} + \alpha \mathcal{L}_{gcl} + \beta (\mathcal{L}_{uniform}^d + \mathcal{L}_{uniform}^t) \quad (10)$$

The uniformity of the representation is helpful to the GraphCL-DTA model. Firstly, it can be used as a regular term in the loss function to improve the model's generalization performance. Secondly, the uniformity of representations improves the model as the low similarity between the representations of different drugs makes each representation more discriminative.

III. EXPERIMENTS AND DISCUSSION

A. Datasets

TABLE I
THE STATISTICS OF DATASETS

	Davis	KIBA
Targets	442	229
Drugs	68	2111
Interactions	30056	118254
Training samples	25046	98545
Test samples	5010	19709

In this section, we use two mainstream datasets, Davis [27] and KIBA [28], to evaluate the proposed model. As shown in Table 1, the Davis dataset contains 442 proteins, 68 drugs, and 30,056 drug-target binding affinity values, where the values range from 5 to 10.8. The KIBA dataset contains 229 proteins, 2111 drugs, and 118254 drug-target binding affinity values, ranging from 0 to 17.2.

B. Experimental settings and evaluation metrics

Consistent with the settings of models such as GraphDTA, we divide all drug-target binding affinity values in the dataset into six equal parts, one of which is used as the test set. The rest uses 5-fold cross-validation to train the model and determine hyperparameters. We measure model performance using three mainstream evaluation metrics: Mean Square Error (MSE), Concordance Index (CI), and r_m^2 . The smaller the value of MSE, the better the model's performance. The larger the value of CI and r_m^2 , the better the model's performance. Equations (10)-(13) show the calculation of these metrics.

$$MSE = \frac{1}{N} \sum_i (p_i - \varrho_i)^2 \quad (11)$$

$$CI = \frac{1}{T} \sum_{\varrho_i > \varrho_j} \mathcal{H}(p_i - p_j) \quad (12)$$

$$\mathcal{H}(x) = \begin{cases} 1, & x > 0 \\ 0.5, & x = 0 \\ 0, & x < 0 \end{cases} \quad (13)$$

$$r_m^2 = r^2 \times (1 - \sqrt{r^2 - r_0^2}) \quad (14)$$

p is the prediction result corresponding to binding affinity ϱ . T is a regularization term. $\mathcal{H}$ is a piecewise function. r^2 represents the square correlation coefficient, and the difference between r_0^2 and the former is that its intercept is 0.

TABLE II
THE SETTING OF HYPERPARAMETERS

Hyperparameters	Change interval	Default
Learning rate	[0.0001, 0.0005, 0.001, 0.005]	0.0005
Batch size	[128, 256, 512, 1024]	512
GCN layers	[1, 2, 3, 4, 5]	3
Drug representation dimension	[64, 128, 256, 512]	128
Target representation dimension	[64, 128, 256, 512]	128
Contrastive loss weight α	[0, 0.01, 0.1, 0.5, 1]	0.5
Uniformity loss weight β	[0, 0.01, 0.1, 0.5, 1]	0.5

C. Hyperparameter settings

Table 2 shows the hyperparameters and their default values in the GraphCL-DTA model. The specific settings are as follows. The change interval of the learning rate is [0.001, 0.005, 0.01, 0.05], and the change interval of the batch size of the training set and the test set is [128, 256, 512, 1024]. The number of GCN layers changes in the interval [1, 2, 3, 4, 5]. The change intervals of the drug and target representation dimensions are [64, 128, 256, 512]. The change intervals of the

contrastive loss function weight and the uniformity loss weight are both $[0, 0.01, 0.1, 0.5, 1]$. Appropriate hyperparameter values are determined from these intervals through the validation set. The default values of the above hyperparameters are 0.005, 512, 3, 128, 128, 0.5 and 0.5.

D. Effectiveness of graph contrastive learning

In this work, we design a graph contrastive learning framework for molecular graphs to learn drug representations. By controlling the magnitude of random noise, the semantics of molecular graphs are better preserved. The graph contrastive framework can learn essential and effective drug representations without additional supervised data. To show whether this graph contrastive learning framework improves the model's performance, we compare it to the model described below. The first comparison model is GCN-DTA. This model is equivalent to the GraphCL-DTA model without the graph contrastive learning framework, which directly uses the GCN to learn the representation of drugs. The second comparison model is GCL-DAT, which replaces the graph contrastive learning in the GraphCL-DTA model with the dropout-based graph contrastive learning. This comparison aims to observe the impact of the semantics of molecular graphs on model performance.

The experimental results of the above models on the Davis and KIBA datasets are shown in Table 3. Figure 4 shows the GraphCL-DTA model's relative improvement percentage under three evaluation metrics (Compared with the GCN-DTA and GCL-DTA models). Firstly, by comparing the GraphCL-DTA model with the GCN-DTA model, we find that the performance of the GraphCL-DTA model has been improved with the support of the graph contrastive learning framework. Compared with the GCN-DTA model, the three metrics of the GraphCL-DTA model on the Davis dataset have been relatively improved by 3.0%, 0.5%, and 3.1%, respectively. The three metrics on the KIBA dataset have relatively increased by 0.7%, 0.1%, and 1.0%, respectively. It is worth noting that the GraphCL-DTA model has a noticeable effect on the Davis dataset while showing a slight relative improvement effect on the KIBA dataset. This is mainly because the Davis dataset has less supervised data, which hinders drug representation learning. Without relying on supervised data, graph contrastive learning learns better drug representation using drug molecular graphs for self-supervised learning, thereby improving model performance. The KIBA dataset has more supervised data, as a result of which, the relative improvement effect of the graph contrastive learning is limited. However, the upper limit of the model's performance can still be improved to a certain extent.

In addition, on the Davis dataset, the MSE value and CI value of the GCL-DTA model are better than those of the GCN-DTA model. This shows that graph contrastive learning based on dropout can improve the model's performance in the Davis dataset with less labeled data. However, on the KIBA dataset with more labeled data, the performance of the GCL-DTA model is worse than the GCN-DTA model. This is mainly because the GCL-DTA model uses the dropout method to generate contrastive views, which destroys the semantic information of molecules to a certain extent, so the model learns the wrong drug features.

Most importantly, compared with the GCL-DTA model, the MSE and r_m^2 values of the GraphCL-DTA model on the Davis dataset have relatively increased by 1.1% and 4.1%, respectively. The three metrics on the KIBA dataset have relatively increased by 2.1%, 0.7%, and 3.2%, respectively. This shows that compared with the dropout-based graph contrastive learning, the graph contrastive framework in the GraphCL-DTA model can better retain the semantic information of molecule graphs and does not need to design a complex data augmentation strategy, positively affecting model performance. The discussion above validates the effectiveness of the graph contrastive learning framework proposed in this work for molecular graphs.

E. Effectiveness of optimizing for representation uniformity

In this work, we design a new loss function that can be directly used to adjust the uniformity of drug representation and target representation smoothly. Uniformity of representation is a property defined in recommender systems to measure the quality of representation and was later found helpful for improving model performance. To explore whether this loss function optimized for representation uniformity helps improve the performance of the GraphCL-DTA model, we run the following experiments. We change the value of the uniformity loss weight β in the interval $[0, 0.1, 0.5]$ to observe the impact of different uniformity losses on the model performance.

Table 4 shows the performance of the GraphCL-DTA model under different uniformity loss weights β . Figure 5 shows the relative improvement percentage of the GraphCL-DTA model under three evaluation metrics. (Compared with the GraphCL-DTA model when $\beta=0$). Firstly, when the value of the uniformity loss β is equal to 0, the uniformity is not optimized, and the prediction performance of the GraphCL-DTA model on both the Davis and KIBA datasets is not the best.

Compared with the GraphCL-DTA model when the value of β is 0, the above results are relatively improved by 1.8% on the Davis dataset and 3.2% on the KIBA dataset. In addition, the sensitivity of Davis and KIBA datasets to β is different. The GraphCL-DTA model achieves the best performance on the Davis dataset when the uniformity loss weight β is 0.1. While on the KIBA dataset, the value of β is 0.5. This may be caused by the different data scales of the two datasets.

Therefore, the above discussion shows that directly optimizing the uniformity of drug representation and target representation can distribute the representation of drugs or targets evenly on the sphere, improving the representation quality of drugs and targets, thereby enhancing the predictive ability of the GraphCL-DTA model.

F. Comparison experiment

We compare the GraphCL-DTA model with recent drug-target binding affinity prediction methods to verify the former's superiority. These state-of-the-art methods are described below.

- **DeepDTA** [11]: DeepDTA inputs the SMILES string of the drug and the protein sequence of the target into the

TABLE III
THE EXPERIMENTAL RESULTS OF THE GCN-DTA, GCL-DTA AND GRAPHCL-DTA ON THE DAVIS AND KIBA DATASETS

Dataset	Methods	MSE	CI	r_m^2
Davis	GCN-DTA	0.260	0.885	0.662
	GCL-DTA	0.255	0.892	0.649
	GraphCL-DTA	0.252	0.890	0.676
KIBA	GCN-DTA	0.140	0.887	0.786
	GCL-DTA	0.142	0.881	0.769
	GraphCL-DTA	0.139	0.888	0.794

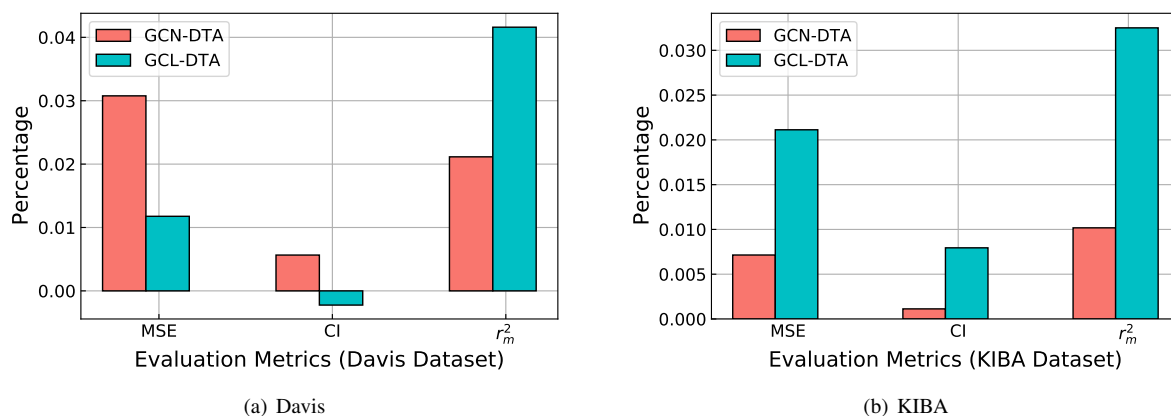


Fig. 4. The relative improvement percentage of the GraphCL-DTA model under three evaluation metrics. (Compared with the GCN-DTA and GCL-DTA models)

TABLE IV
THE PERFORMANCE OF THE GRAPHCL-DTA MODEL UNDER DIFFERENT UNIFORMITY LOSS WEIGHTS

Dataset	Weight β	MSE	CI	r_m^2
Davis	value=0	0.260	0.885	0.662
	value=0.1	0.252	0.889	0.675
	value=0.5	0.261	0.882	0.667
KIBA	value=0	0.140	0.887	0.786
	value=0.1	0.134	0.893	0.787
	value=0.5	0.130	0.894	0.801

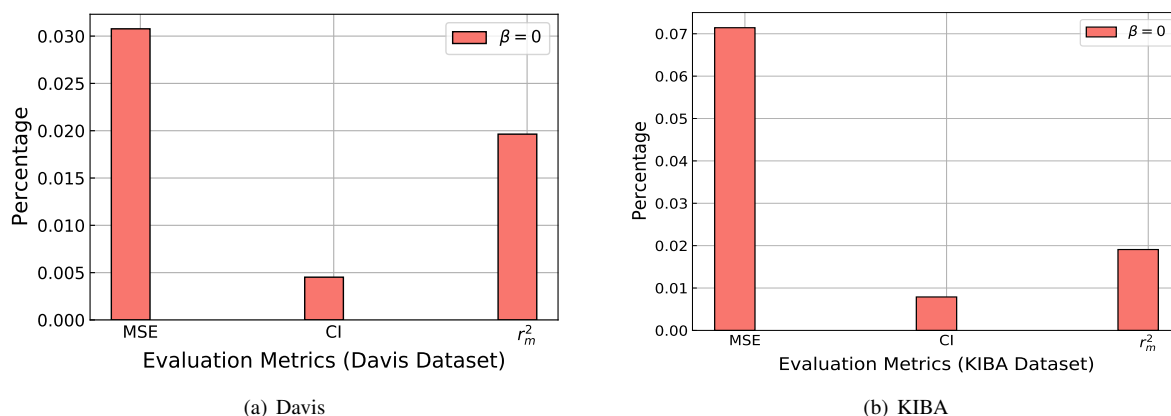


Fig. 5. The relative improvement percentage of the GraphCL-DTA model under three evaluation metrics. (Compared with the GraphCL-DTA model when $\beta=0$)

convolutional neural network to learn the representation of the drug and the target. Then, DeepDTA inputs the concatenated drug representation and target representation

into a fully connected neural network to derive predicted binding affinities.

- **WideDTA** [7]: Based on DeepDTA, WideDTA intro-

TABLE V
THE PERFORMANCE OF THE GRAPHCL-DTA MODEL WITH STATE-OF-THE-ART MODELS

Dataset	Methods	MSE	CI	r_m^2
Davis	DeepDTA	0.261	0.878	0.630
	WideDTA	0.262	0.886	0.633
	CPInformer	0.277	0.874	0.618
	GANsDTA	0.276	0.881	0.653
	GraphDTA	0.260	0.885	0.662
	GraphCL-DTA	0.248	0.890	0.683
KIBA	DeepDTA	0.194	0.863	0.673
	WideDTA	0.179	0.875	0.675
	CPInformer	0.183	0.867	0.677
	GANsDTA	0.224	0.866	0.775
	GraphDTA	0.140	0.887	0.767
	GraphCL-DTA	0.129	0.895	0.805

duces more chemical and biological text information to strengthen the representation of drugs and targets.

- **GraphDTA** [14]: The GraphDTA model uses molecular graphs and graph neural networks to learn the representation of drugs, and uses protein sequences and 1D convolutional neural networks to learn the representation of targets. The prediction module uses a multi-layer fully connected neural network.
- **CPInformer** [29]: The novelty of the CPInformer model is that it uses structural graph features and functional class fingerprints to extract drug representations to distinguish drugs with similar structures. The encoder with dense connections is used to learn multi-scale protein representations.
- **GANsDTA** [30]: The GANsDTA model uses a semi-supervised generative adversarial network as the representation learning module, which can learn representations from labeled and unlabeled data.

Table 4 shows the experimental results of the GraphCL-DTA model and all comparison models. It is obvious that with the support of graph contrastive learning and representation uniformity, the GraphCL-DTA model has achieved the best performance in all datasets and evaluation metrics.

Firstly, the GraphDTA model is currently the most advanced one, and the input of GraphDTA is consistent with the GraphCL-DTA model. The GraphDTA model can be regarded as a simplified version of the GraphCL-DTA model, which removes graph contrastive learning and representation uniformity. Compared with GraphDTA, the relative improvement of the GraphCL-DTA model on the two datasets is 2.7% and 4.5%. Such relative improvement indicates that the graph contrastive learning and representation uniformity greatly help the model's predictive ability. Since the GraphDTA model does not utilize the structure of the molecular graph of the drug itself, it leads to its poor performance. The GraphCL-DTA model considers the drug's molecular structure and uses graph contrastive learning to mine the internal information of the structure, the performance of which is better as a result.

In addition, the DeepDTA, WideDTA, and GANsDTA models that use the 1D structure of the drug as input perform worse than the GraphDTA and GraphCL-DTA models that use the 2D structure of the drug as input. This shows that the

information in the molecular graph structure is greater than that of the SMILES string. Moreover, the original purpose of the CPInformer model is to predict the association between compounds and proteins, so its prediction effect is not good in drug-target binding affinity tasks.

The analysis and discussion of the above experimental results prove that the GraphCL-DTA model performs better than the state-of-the-art models, thus illustrating the excellence of the model proposed in this work. There are two main reasons. Firstly, the graph contrastive learning framework reduces its dependence on supervised data, which helps learn the more essential and effective drug representation. Secondly, the loss function optimized for the uniformity of drug representation and target representation improves the quality of representation.

IV. CONCLUSION

In this work, we propose GraphCL-DTA for drug-target binding affinity prediction, which can better preserve the semantic information of molecular graphs. The more essential and effective drug representation can be learned through this graph contrastive framework without additional supervised data. This way, the dependence on wet experiment data can be reduced, resulting in significant cost savings. In addition, we propose a loss function optimized for the uniformity of drug representation and target representation, which can be used to improve the quality of representation directly. Extensive experiments on two public datasets demonstrate the effectiveness of the graph contrastive learning framework and uniformity loss function. Moreover, the GraphCL-DTA model is verified to outperform five comparison models. However, the GraphCL-DTA model only considers the features of atoms when extracting the drug representation and does not consider the features of the bonds between atoms, which loses the information contained in part of the molecular graph. Besides, the 1D CNN cannot capture the correlation information between protein sequences. In future work, we will develop GCN models considering atomic and bond features in molecular graphs. More advanced sequence models, such as a transformer, will be used to extract a better representation of the target. In addition, knowledge graphs can be used in related fields [31], [32] to overcome data sparsity and enhance interpretability. In future work, we will study how to introduce knowledge graphs

to the drug-target binding affinity prediction task. Similarly, the drug-target binding affinity prediction task can be presented in the form of a graph, and research tasks in other graph fields [33] can be used to improve the GraphCL-DTA model. Finally, recent work on drug-target binding affinity prediction [34], [35] also provides ideas for future improvements.

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